

AP6XXX 12x12mm modules_General Spec.

Module	Wi-Fi IF	BT IF	Audio	Wi-Fi Spec.	BT Spec.	Ref Clock	VDDIO
AP6281S	SDIO 2.0/3.0	UART	PCM	2.4/5/6e GHz A/B/G/N/AC/AX	5.3	59.97MHz	1.8V

AP6XXX_12x12mm_modules_Ref. Design_Key Component Spec.

X'tal Spec. (Y3)	
59.97MHz	Part Number : Hosonic ETSB59E0X1L00E Operation Freq. : 59.97MHz Freq. Tolerance : -7 to +7ppm (25 +/- 3) Operation Temp. : -30 to +85 Freq. Stability : -10 to +10ppm Load Cap. : 8pF Size : 1.6mm x 1.2mm

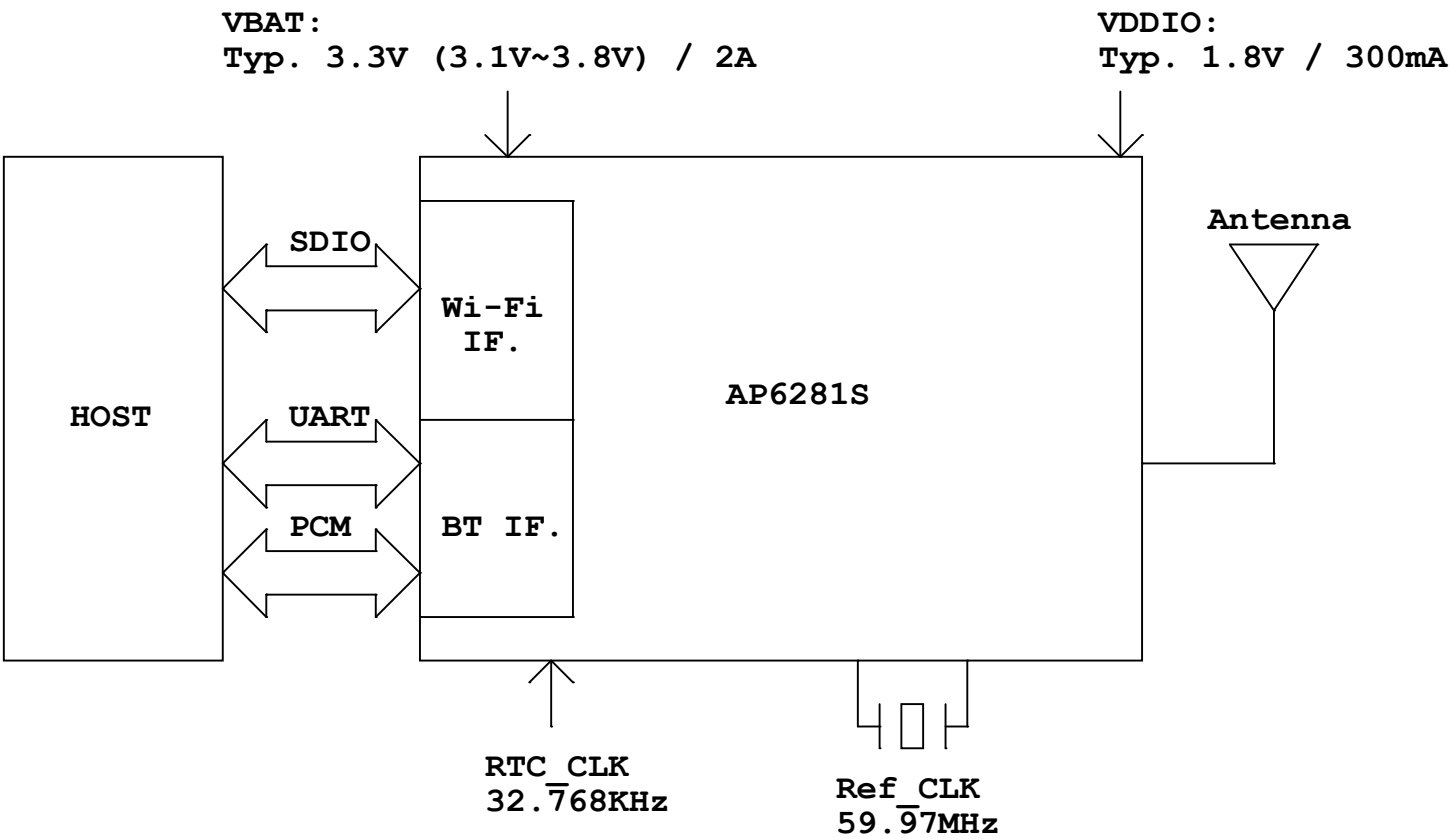
Crystal Oscillator1 Spec. (X1)	
32.768KHz	Part Number : YOKE SO3225-000032-FXA-FBE-AA Operation Freq. : 32.768 KHz Operation Temp. : -40 to +85 Freq. Stability : -25 to +25ppm Size : 3.2mm x 2.5mm

Power Inductor Spec. (L3)	
2.2uH	Part Number : Murata DFE201610E-2R2M-P2 Nominal Value : 2.2uH Tolerance : +/-20% DC Resistance : 140 m (max.) Rated Current based on : 2.4A(max.) Inductance Change Rated Current based on : 1.7A(max.) Temperature Change

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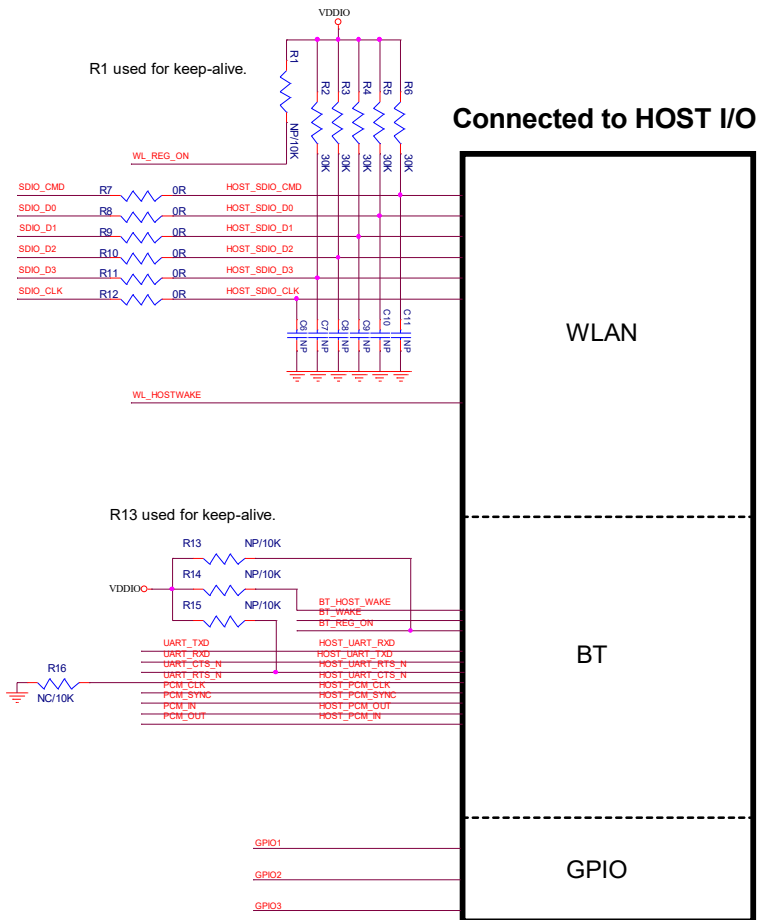
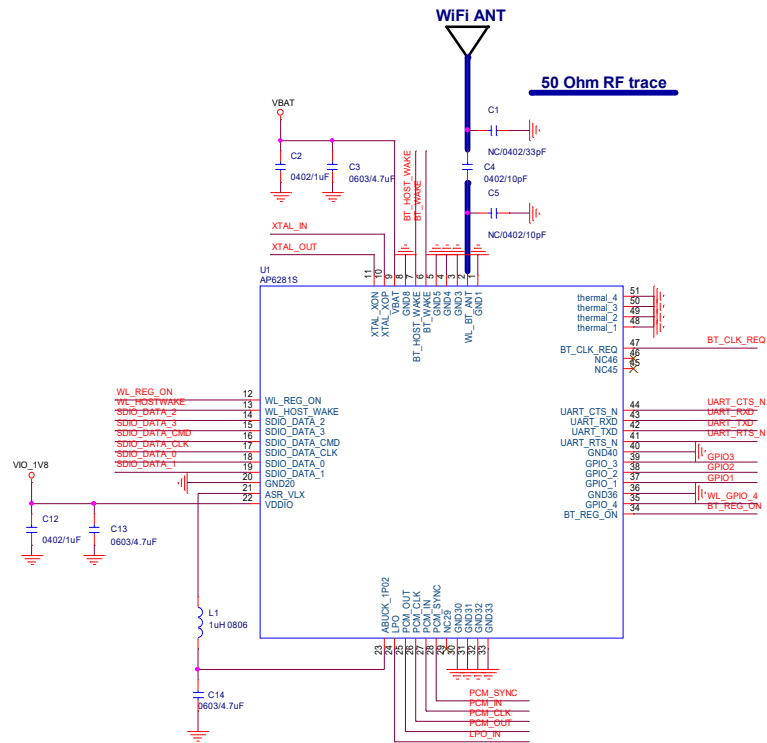
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AP6XXX_12x12mm_Block Diagram

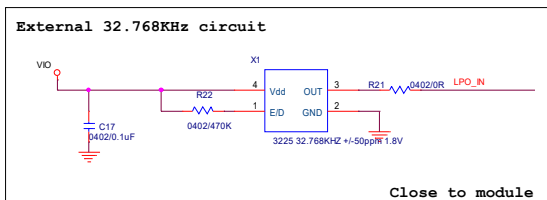
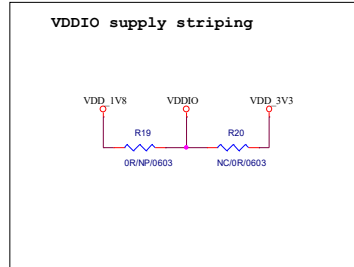
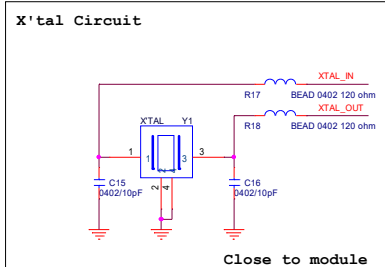


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AP6281S Reference Design Circuit



VBAT:3.3V
VIO:1.8V



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SDIO 2.0 / 3.0 Switch

	R19	R20
SDIO2.0 VDDIO=3.3V	NP	0 ohm
SDIO3.0 VDDIO=1.8V	0 ohm	NP

X'TAL(Y1) circuit reference number
Y1 = Hosonic ETSB59E0X1L00E

	C15	C16	R17	R18
59.97MHz CL=7pF	10pF	10pF	Bead 120 ohm @100Hz	Bead 120 ohm @100Hz

C19 C20 R12 R13 could be changed and optimized
as diverse main board designs and X'tal vendors.

Please contact X'tal vendor for technical support
and evaluation.

